

Slides 1.9: The Six Nutrients

Lesson 1.9, one day. Thirteen slides. Build the deck from this outline; keep each slide to the lines shown. Student-facing text reads at grade 6 to 7.

Slide 1: Do now

- Write what you ate for breakfast today, or the last thing you ate.
- Next to each food, write what you think it did for your body.
- One word is fine: energy, muscle, bones, nothing.
- Two minutes. Pencils down when the timer sounds. Notes: Cold call three students, write their foods in a column on the board. Do not correct anything yet.

Slide 2: You already know some of this

- On the pre-assessment, [Sal: number] of [Sal: number] of you named at least five nutrients.
- So today we skip what you know.
- We get to what your body DOES with each one.
- And why your little cousin and your grandmother do not eat like you. Notes: Fill the numbers from the Lesson 0.5 tally. If two thirds or more named five nutrients, jump from this slide to Slide 9 (the chart) and go straight to the sort.

Slide 3: Every food is a delivery truck

- The truck is the food. The delivery is the nutrient.
- Six kinds of delivery. Six nutrients.
- Carbohydrates, proteins, fats, vitamins, minerals, water.
- Memory line: carbs for go, protein for grow, fat for stored go and brain, vitamins and minerals run the show, water carries it all. Image: six small delivery trucks in a row, each with one nutrient name on the side.

Slide 4: Carbohydrates

- Job: your main quick energy. Your brain runs on it.
- Foods: bread, rice, pasta, oatmeal, potatoes, fruit, beans.
- Better versions: whole grains and fruit. They bring fiber along.
- Not enough: tired, foggy, cranky. Image: a bowl of rice, a slice of whole wheat bread, a banana.

Slide 5: Proteins

- Job: build and repair. Muscle, skin, hair, blood.
- Foods: chicken, fish, eggs, beans, milk, cheese, yogurt, tofu, nuts.
- You are growing right now. This is the nutrient that does it.
- Not enough: slow healing, weak muscles. Image: eggs, a piece of grilled chicken, a bowl of black beans.

Slide 6: Fats

- Job: stored energy, protects your organs, keeps you warm, carries vitamins A, D, E, and K.
- Foods: oils, butter, avocado, nuts, peanut butter, salmon, cheese.
- A toddler's brain is mostly built from fat. Little kids need more of it than you do.
- Too much of the wrong kind: heart trouble later. Saturated fat is the "get less" one on a label. Image: an avocado cut in half, a bottle of olive oil, a handful of almonds.

Slide 7: Vitamins

- Job: run the body's processes. Seeing in the dark (A), healing cuts (C), strong bones (D), using energy (B).
- Foods: fruits and vegetables most of all; milk for D; eggs; whole grains for B.
- You need tiny amounts. Missing one for a long time causes real problems.
- Sailors with no vitamin C got scurvy. Limes fixed it. Image: a plate of broccoli, carrots, and strawberries.

Slide 8: Minerals and water

- Minerals: build bones and teeth (calcium), carry oxygen in the blood (iron), balance fluids (potassium, sodium).
- Mineral foods: milk, cheese, yogurt, spinach, beans, meat, bananas.
- Water: carries nutrients in and waste out; controls your temperature; more than half of you is water.
- Water foods: water, milk, soup, watermelon, cucumber. Thirst means you are already behind. Image: a glass of milk next to a glass of water, a wedge of watermelon.

Slide 9: The chart

Nutrient	Job	Two foods
Carbohydrates	quick energy	
Proteins	build and repair	
Fats	stored energy, brain, carries vitamins	
Vitamins	run the processes	
Minerals	bones, blood, fluids	
Water	carries everything	

Notes: Leave this on the screen during the sort. Students fill the foods column on the handout.

Slide 10: The Nutrient Sort

- 30 food cards. Six piles.

- Every card goes in a pile.
- If a card fits two piles, put it between them and write why.
- Nine minutes. I am not telling you if you are right. Ask “what does that food do for me?” Image: six pile cards fanned out with a food card between two of them. Notes: Circulate with the clipboard checklist: water and vitamins piles have three or more cards; team can explain a card without reading it; team found a two-pile card.

Slide 11: Three people, three bodies

- A two year old learning to walk and talk.
- A 14 year old on a travel soccer team, five practices a week.
- A 78 year old who walks a mile most days and eats less than they used to.
- For each: the TWO nutrients they need most and one sentence why, using a job from the chart. Image: three simple figures of different heights side by side, no faces needed. Notes: After five minutes, show the answer on the next slide only after two teams have shared per person.

Slide 12: What the science says

- Toddler: fats (brain), calcium and iron (bones and blood), protein. Small stomach, small meals often.
- Teen athlete: carbohydrates (energy), water (sweat), protein (muscle during a growth spurt), calcium and iron.
- Grandparent: protein (keeps muscle), calcium and vitamin D (slows bone loss), water (thirst signal weakens), fiber. Fewer calories, not fewer nutrients.
- Same six nutrients. Different amounts. That is the whole lesson. Notes: Push back once: “The grandparent still needs protein. Why?” Source: Dietary Guidelines for Americans, life stage chapters. *[Sal: check the edition.]*

Slide 13: Exit card

- 3 nutrients and one job each.
- 2 foods you ate this week and the nutrient each one mostly gave you.
- 1 question you still have about what your body needs.
- Three volunteers read one line. Notes: Sort the cards into “got it” and “needs more.” More than a third in “needs more” means a two-minute nutrient review opens Lesson 1.10.